10/12/24 M/S

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[23CSBS303]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER- 2024

OBJECT ORIENTED PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART- A			5 X 2=10
1.	a.	Define Namespace in C++.	L1	CO1	02
	b.	List out the Access Specifires in C++.	L1	CO2	02
	c.	Define Virtual function.	L1	CO3	02
	d.	What is generic programming?	L1	CO4	02
	e.	Define Use case.	L1	CO5	02
		PART-B			**6 X 5=30**
2.	a.	What is Scope and lifetime of a Variable? Explain with an example program.	L2	CO1	05
		OR			
	b.	Briefly explain about Command line arguments and its types.	L2	CO1	05
3.	a.	Write short notes on Data Abstraction and Data Encapsulation.	L1	CO2	05
		OR			
	b.	What is the use of this keyword? Explain with an example program.	L1	CO2	05
4.	a.	Define virtual function. List out the properties of Virtual function.	L2	CO3	05
		OR			
	b.	Differentiate function overloading and function overriding in c++.	L4	CO3	05
5.	a.	Briefly discuss about formatted i/o functions in c++.	L2	CO4	05
		OR			
	b.	Write short notes on i/o streams in c++	L2	CO4	05
6.	a.	Design a use case diagram for Hospital Management systems.	L3	CO5	05
		OR			
	b.	Illustrate Sequence diagram. Draw the sequence diagram for Hotel Reservation process.	L3	CO5	05
7.	a.	Describe the uses of new and delete operator in c++ with an example.	L2	CO1	05
		OR			

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	b.	Define class. Explain with an example how the member functions of the class can be defined?	L2	CO1	05
		PART-C			**6 X 10=60**
8.	a.	Explain Data type. Explain different data types in C++ with examples.	L2	CO1	10
		OR			
	b.	Explain Call by value and Call by reference with an example c++ program.	L2	CO1	10
9.	a.	Explain friend function with an example.	L2	CO2	10
		OR			
	b.	Explain in detail about different types of constructors with example c++ code.	L2	CO2	10
10.	a.	Define Operator Overloading? Explain it with an example C++ program.	L2	CO3	10
		OR			
	b.	Define inheritance. Explain any two types of inheritance with sample code.	L2	CO3	10
11.	a.	Illustrate the concept of function template with the C++ program.	L3	CO4	10
		OR			
	b.	Illustrate file handling in c++ With the sample code.	L3	CO4	10
12.	a.	What is meant by UML? Discuss about various diagrams of UML.	L2	CO5	10
		OR			
	b.	Explain the steps involved in converting UML diagrams into c++ code.	L2	CO5	10
13.	a.	Illustrate control flow statements. Explain each with an example c++ code.	L3	CO1	10
		OR			
	b.	Write a C++ program to create a class called Car that has private member variables for company, model, and year. Implement member functions to get and set these variables.	L3	CO1	10